Amazon Logistics

Data Analysis
Case Study

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Problem

Amazon's logistics sector has been experiencing a critical decline in on-time delivery performance, directly impacting customer experience and satisfaction.

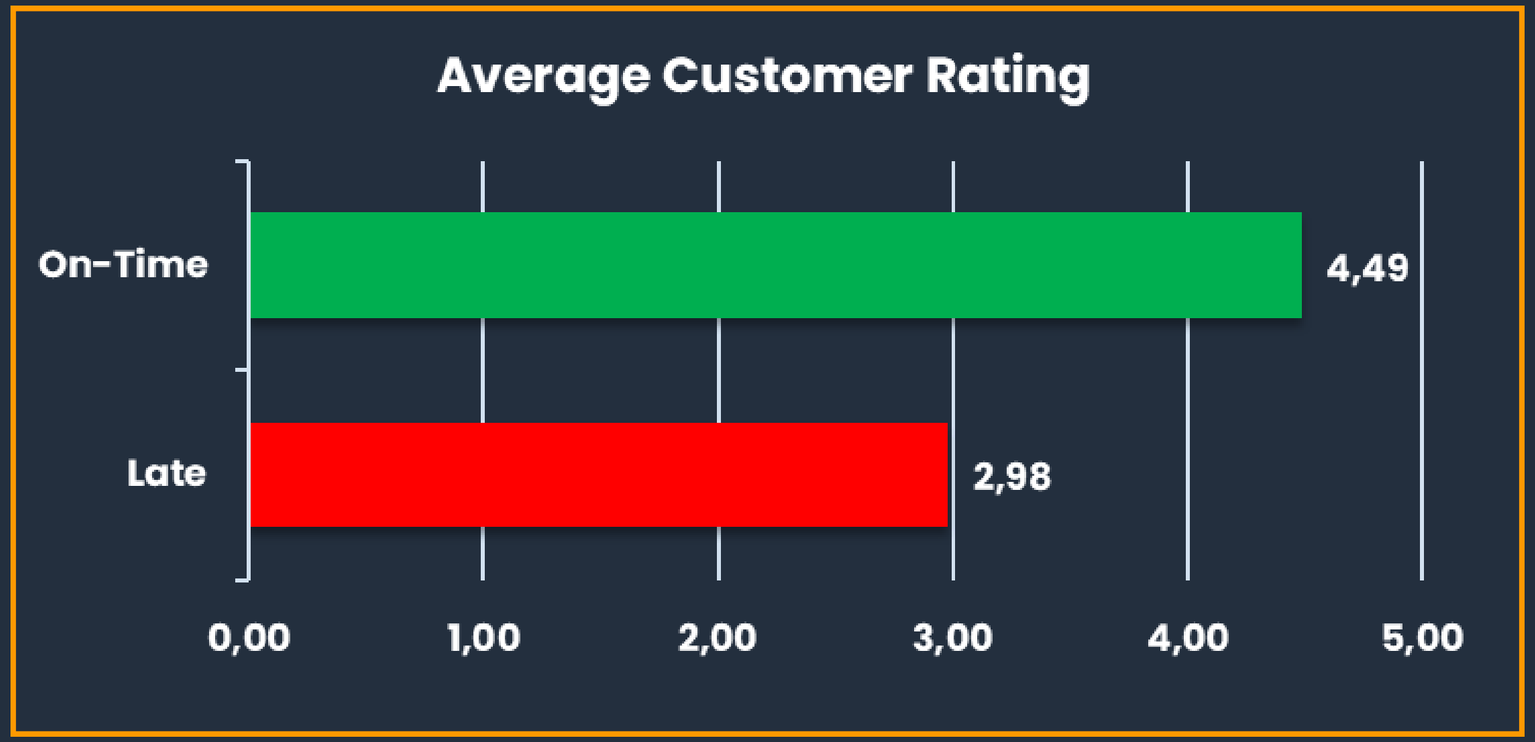
KPI	Current	Target
OTIF	61%	>= 97,0%
Cost per Order	$ 7,25	<= $ 6,20
Customer Rating	3,91	>= 4,50
Late Deliveries	38,1 %	<= 3,00%

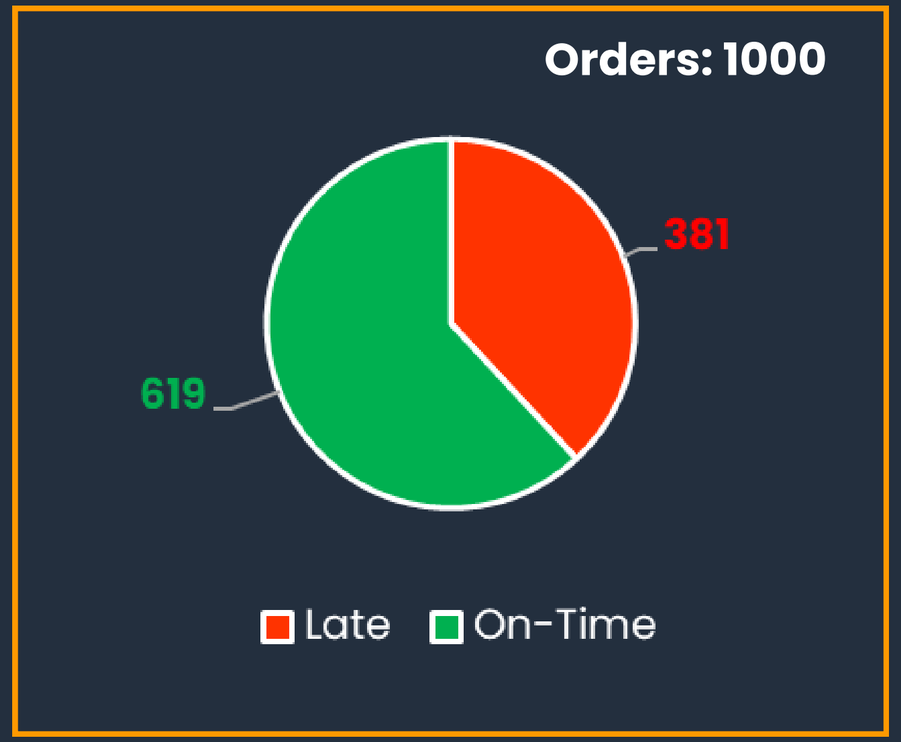


All KPIs are below target, indicating a systemic operational failure.

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Delays vs. Customer Experience





Late deliveries cause a **1.51-point drop in average** customer rating, from 4.49 to 2.98. **With 38.1% of orders arriving after the promised deadline**, customer experience is being severely impacted.

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Carriers

Carrier	Delivered On-Time		Orders	OTIF	Avg. Shipp. Cost	Avg. Cust. Rating
	No	Yes				
BudgetExpress	150	155	305	**50,82%**	**$ 7,24**	**3,78**
FastShip	123	269	392	68,62%	**$ 7,23**	3,99
QuickLog	108	195	303	64,36%	$ 7,29	3,94

BudgetExpress accounts for the **highest volume of delays (150 orders) with an OTIF of only 50.82% and a rating of 3.78**. This is not a cost-benefit trade-off: the shipping cost per order ($7.24) is equivalent to FastShip's ($7.23), which operates with 35% fewer delays.



Region

Region	Delivered On-Time		Orders	OTIF	Avg. Shipp. Cost	Avg. Cust. Rating
	No	Yes				
East	89	165	254	64,96%	$ 7,38	3,91
North	**104**	137	241	**56,85%**	$ 7,17	**3,85**
South	102	154	256	60,16%	$ 7,22	3,88
West	86	163	249	65,46%	$ 7,23	4,01

The **North region leads in delays with only 56.85% of on-time deliveries and an average rating of 3.85**, the worst indicators across all regions. Despite having the lowest average shipping cost, the issue is operational, stemming from either carrier quality or fulfillment center efficiency within the region.



Fullfilment Centers

FC	Delivered On-Time		Total	OTIF
	No	Yes		
FC-A	91	216	307	70,36%
FC-B	86	193	279	69,18%
FC-C	204	210	**414**	**50,72%**

FC-C processes 41% of all orders (414 out of 1,000) and has the **worst OTIF in the operation (50.72%)**. Alone, it accounts for **204 out of 381 total delays, more than half of all Amazon's delays.** The high volume combined with low efficiency confirms a critical operational bottleneck.



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Carriers vs. Regions

OTIF (On-Time in Full Delivery)			
Região	**BudgetExpress**	FastShip	QuickLog
North	**51,39%**	**61,22%**	**56,34%**
South	47,06%	69,77%	63,53%
East	55,29%	71,57%	67,16%
West	49,21%	71,70%	70,00%

Cross-referencing carriers and regions confirms that **BudgetExpress has the worst OTIF across all regions, with the highest rate of late deliveries.** Additionally, **the North region is the most critical for all carriers.** The worst scenario is BudgetExpress in the South, with only 47.06% on-time delivery (OTIF).

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Customer: Prime vs. Standard

Type	Delivered On-Time		Orders	OTIF	Avg. Shipp. Cost	Avg. Cust. Rating
	No	Yes				
Prime	232	398	630	63,17%	$ 8,03	**3,94**
Standard	149	221	370	59,73%	$ 5,93	3,86

Prime customers represent 63% of all orders and pay an average of $8.03 per order, 35% more than Standard customers ($5.93). Despite the premium price, Prime OTIF is only 63.17%, nearly identical to Standard (59.73%). Customers paying more are receiving the same poor service level, resulting in an average rating of 3.94. **This represents a direct strategic risk: dissatisfaction may lead to Prime subscription cancellations and loss of recurring revenue.**



Solutions

1

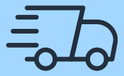
Carrier Management

Problem: BudgetExpress

- High Delay Rate
- Similar cost to FastShip (Better OTIF)

Solutions:

- Reduce BudgetExpress delivery volume.
- Transfer orders to other carrier with better cost and fewer delays.



2

FC-C Optimization

Problem: FC-C

- Process 41% of orders.
- Responsible for 53,5% of all delays.

Solutions:

- Redistribute volume to FC-A and FC-B.
- Study and expand FC-C capacity.
- New FC-D.



3

Prime First

Problem: Prime Customers

- Pay 35% more, and receive the same service quality as Standard.

Solutions:

- Prioritaze Prime deliveries.
- Differentiated SLA segmentation for Prime
- Offer compensation for delayed Prime orders (discounts).



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OTIF
61.90%
Target >= 97,00%

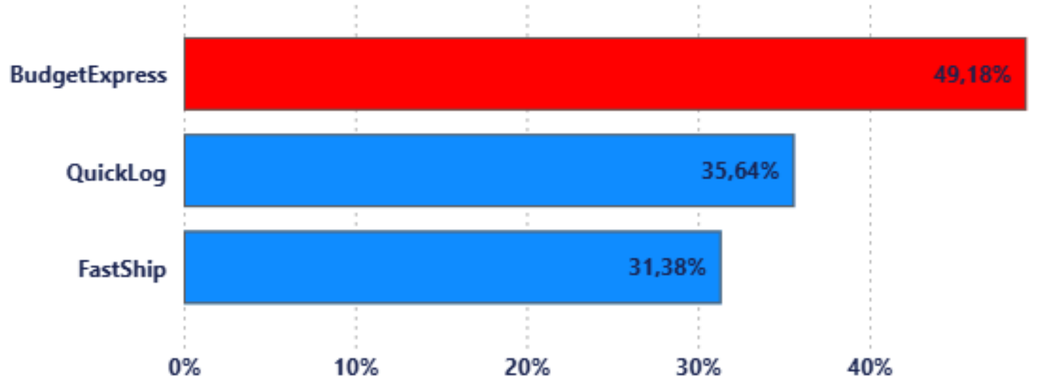
Cost Per Shipping
$7.25
Target: $6,20

Late Deliveries
38.10%
Target < 3,00%

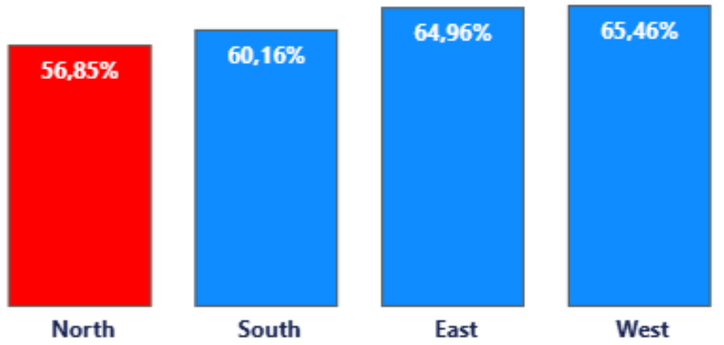
Customer Rating
3.91
Target > 4,50

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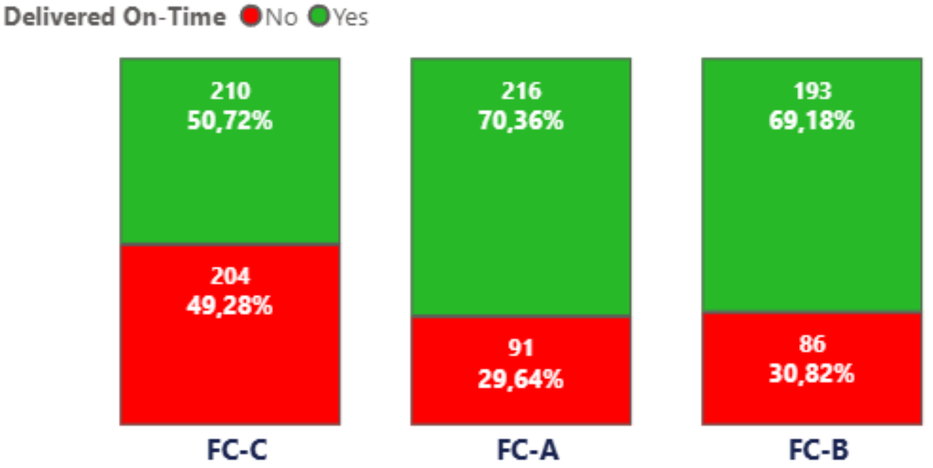
Late deliveries by Carrier



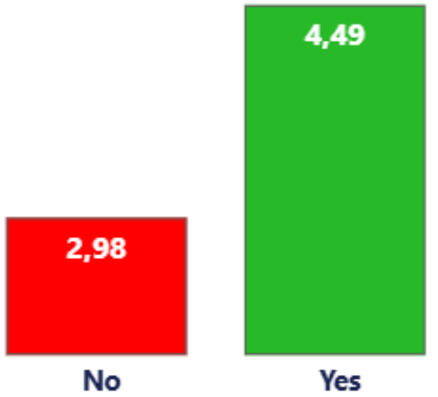
OTIF by Region



Orders by Fullfliment Center



Customer Rating vs. Delivery On-Time



Carrier
- [x] BudgetExpress
- [x] FastShip
- [x] QuickLog

Region
- [x] East
- [x] North
- [x] South

Fullfilment Center
- [x] FC-A
- [x] FC-B
- [x] FC-C

Customer Type
Prime
Standard

Thank you!

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